water & sanitation

Department:
Water and Sanitation
REPUBLIC OF SOUTH AFRICA

SITA

Business Requirements:

IT Asset Management System

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1.Document Control

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1.2 Version History

Version	Author	Version Information
1.0	Siphesihle Majola	Draft
1.1	Siphesihle Majola	Draft
2.0	Siphesihle Majola	Final
3.1		Draft

2. Background

2.1 Introduction

The IT Asset Management System (ITAMS) is a .NET (C#) MVC web application designed to capture, manage, and report on IT assets such as computers, servers, software applications, and users within an organization's IT infrastructure. The system stores asset details, records status, tracks assignments, manages user access, and provides comprehensive reporting for management purposes.

2.2 Evolution & Context

Previously, IT assets were tracked using spreadsheets and manual reporting. ITAMS introduces a centralized system for recording asset information and generating reports efficiently.

2.3 Benefits & Challenges

Benefits:

- Centralized visibility of IT assets
- Simplified reporting and auditing
- Accurate status tracking of servers, applications, computers, and users
- Role-based access control for enhanced security
- Automated password management and expiration
- Asset assignment tracking and dependency management
- Comprehensive audit logging
- IT Support ticket management for issue tracking and resolution
- Bulk operations (select all, bulk edit, bulk delete) for assets and users
- CSV bulk import for assets and users with validation and result reporting
- License management and expiry tracking for software applications
- PDF and CSV export across all reports and asset lists

Challenges:

- Ensuring data consistency and accuracy
- User training in system usage
- Managing password expiration policies

2.4 Statement of Business Requirement

The system must allow authorized users to:

- Register and track computers, servers, software applications, and users
- Record hardware and software properties and asset ownership
- Track the status of assets and their assignments to users
- Manage user accounts with role-based permissions
- Generate reports and dashboards for IT management
- Ensure role-based access for security
- Manage passwords with expiration policies
- View and update personal profiles
- Track audit logs of all system activities
- Log, track, and resolve IT support tickets linked to assets
- Bulk import assets and users via CSV with template downloads
- Perform bulk operations on assets, users, and support tickets

2.5 Impacted Business Areas

- IT Operations
- System Administration
- IT Audit
- User Management
- Assets
- Desktop Support (Future Enhancement)

3. Scope

In Scope:

- ASP.NET Core MVC application & SQL Server database
- CRUD operations for IT assets (computers, servers, software applications)
- User management with role-based access control & Profile management
- Asset assignment and tracking
- Dashboards, reporting & CSV export functionality

- Password management and expiration
- Audit logging & Asset dependency management
- IT Support ticket system (create, assign, track, and resolve tickets)
- CSV import for assets and users with validation and template download
- Bulk operations: select all, bulk edit, bulk delete for assets, users, and tickets
- PDF and CSV export across all modules
- License management and expiry tracking for software applications

Out of Scope:

- Real-time monitoring
- Third-party integrations
- Analytics
- Automated alerts for asset status
- Mobile apps (Planned for future enhancement)
- Email / push notifications for ticket updates (Planned for future enhancement)

4. Business Benefits

- Centralized asset management
- Reduced errors and duplication
- Simplified auditing and reporting
- Improved visibility into asset status
- Enhanced security through role-based access
- Automated password policy enforcement
- Better tracking of asset assignments and dependencies
- Comprehensive audit trail for compliance

5. Strategic Alignment

Supports digital transformation by enhancing IT governance, compliance, and reporting capabilities. The system aligns with organizational security policies through role-based access control and password management.

6. Key Stakeholders

• System Administrators

• IT Managers

• Network Engineers

• Security Officers

• Developers

• End Users (Employees)

• Desktop Support Team (Future)

7. Implemented Features

- Asset Management
- User Management
- Role-Based Access Control
- Password Management
- Profile Management
- Dashboard
- Reporting
- Server Application Management
- Audit Logging
    - IT Support Ticket Management
    - CSV Bulk Import (Assets & Users)
    - Bulk Operations (Select All, Bulk Edit, Bulk Delete)
    - License Management & Expiry Tracking
    - PDF & CSV Export (All Modules)
- Asset Dependencies

8. Functional Requirements

- Authentication & Authorization
- Password Management
- User Management
- Profile Management

- Asset Management
- Asset Dependencies
- Asset Assignment & Visibility
- Server Application Management
- Dashboard & Reporting
- Search & Filter
- Data Export
- Delete Permissions
- Audit Logging

9. Security & Access Control

- Authentication via ASP.NET Identity
- Role-based authorization
- Password Security & Data Protection
- Audit & Compliance
- Access Restrictions

10. Proposed Timelines

Phase	Planned Duration	Expected Period	Status	Key Activities
Requirements & Analysis	1 week	October 2025	Completed	Gather and define system requirements and roles
System Design	1 week	October 2025	Completed	Design database schema, dashboards, and reports
Development & Integration	2 months	Nov 2025 – Dec 2025	Completed	Develop MVC app, integrate database, implement reporting
Testing & Quality Assurance	1 week	Jan 2026	In Progress	Conduct testing, UAT, and security review
Deployment & Training	1 week	Feb 2026	Planned	Deployment to production, provide training

11. Training & Communication

Live training sessions for IT staff and administrators

• Recorded training videos

• User documentation and guides

• Role-specific training materials

• Password policy communication

12. Business Risk & Mitigation

Risk	Mitigation
Data loss	Regular backups and database recovery plan
Unauthorized access	Strong authentication, RBAC, and password policies
Password security breaches	Password expiration policy and strong password requirements
Asset dependency errors	Comprehensive dependency checks before deletion
Incorrect asset assignments	Clear assignment tracking and validation
Low adoption	User training and engagement sessions

13. Future Enhancements

13.1 IT Support Ticket Module (Implemented)
The IT Support ticket module is now fully implemented. Users can report issues with their assigned assets directly through the system.

The module establishes relationships between:
• Asset Owners (users who own/are assigned assets)
• Desktop Support Team (IT support staff who handle asset issues)
• Administrators (who oversee and manage support tickets)

Key Features (Implemented):
• Asset issue reporting by asset owners
• Ticket creation and assignment to desktop support
• Ticket status tracking (Open, In Progress, Resolved, Closed)

• Communication workflow between asset owners and support team
• Admin oversight and ticket management
• Integration with existing asset management system
• Notification system for ticket updates (planned enhancement)
• Asset history tracking through support tickets

Use Case:
1. User (Asset Owner) reports an issue with their assigned computer/server/application
2. System creates a support ticket and assigns it to Desktop Support team
3. Desktop Support team receives notification and updates ticket status
4. Asset Owner can track ticket progress
5. Admin can view all tickets and manage assignments
6. Resolved tickets are linked to asset history

13.2 Future Enhancements

• Email and push notifications for ticket status updates

• Mobile application (iOS and Android)

• Real-time asset monitoring and automated alerts

• Third-party system integrations (e.g. Active Directory, ITSM tools)

• Advanced analytics and reporting dashboards

14. Recommendation & Approval

Role	Name	Signature	Date
Deputy Director Software Development	Mr. Dziedzi Ramulondi		
Software Developer Trainee	Siphesihle Majola		